

Robert M. Gower

Research Scientist, Flatiron Institute

gowerrobert@gmail.com gowerrobert.github.io [Google Scholar](#) [GitHub](#) [LinkedIn](#) Updated May 11, 2026

[modern optimization](#) [adaptive methods](#) [stochastic optimization](#) [machine learning](#) [randomized numerical linear algebra](#)

Profile

I work on the mathematics of modern optimization: why methods such as Adam, Muon, Polyak step-sizes, and non-Euclidean gradient descent work, fail, and sometimes come up with new methods. My current research sits between optimization theory and machine-learning practice, with a focus on simple algorithms that exploit structure. Previously, I worked on randomized numerical linear algebra, automatic differentiation, and quasi-Newton methods.

Appointments and Education

11/2021–present	Research Scientist , Center for Computational Mathematics, Flatiron Institute New York, NY
06/2021–11/2021	Visiting Scientist , Google Research New York, NY
09/2017–06/2021	Assistant Professor in Machine Learning , Telecom Paris, Institut Polytechnique de Paris Paris, France
01/2020–08/2020	Visiting Scientist , Facebook Research New York, NY
08/2016–09/2017	Laureate Fellow , Fondation Sciences Mathématiques de Paris, ENS and Inria Paris, France
09/2012–06/2016	Ph.D. , Optimization and Operations Research, University of Edinburgh Thesis: <i>Sketch and Project: Randomized Iterative Methods for Linear Systems and Inverting Matrices</i>
03/2009–04/2011	M.Sc. , Applied Mathematics, University of Campinas Thesis: <i>Hessian Matrices via Automatic Differentiation</i>
03/2005–03/2009	B.Sc. , Applied and Computational Mathematics, University of Campinas Brazil

Selected Recent News

2026	The Polar Express received the ICLR 2026 Outstanding Paper Honorable Mention and was presented as an oral.
2026	Upcoming talk at the SIAM Conference on Optimization (OP26) , June 2–5, 2026.
2025	In Search of Adam’s Secret Sauce was an oral presentation at NeurIPS 2025.

Prizes, Honors, and Funding

- ICLR 2026 Outstanding Paper Honorable Mention for *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm*.
- NeurIPS 2025 Top Reviewer.
- ICML 2021 Best Reviewer, Top 10%.

- NeurIPS 2020 Top 10% reviewer.
- Thomas Jefferson Fund grant, “Matrix equations for big data compression”, \$20k, 2019.
- CIFRE Facebook Research funding, \$72k, 2019.
- ICML 2019 Top 5% reviewer.
- LabEx Mathématique Hadamard funding, “Scalable stochastic variance reduced gradient methods”, 8k EUR, 2018.
- Leslie Fox Prize for Numerical Analysis, 2nd place, 2017.
- Fondation Sciences Mathématiques de Paris Fellowship at Inria/ENS, 122k EUR, 2016.
- Best Talk Prize, Irish SIAM student meeting, 2014.
- University of Edinburgh Ph.D. scholarship, 85.1k GBP, 2012.

People and Mentoring

- [Nidham Gazagnadou](#), Ph.D. student, 2018–2021; now Research Scientist at Sony AI.
- [Rui Yuan](#), Ph.D. student, 2019–2023; now AI Research Scientist at Stellantis.
- [Samy Jelassi](#), intern, 2017; Ph.D. at Princeton.
- [Si Yi Meng \(Cathy\)](#), intern, 2022 and 2023.
- [Fabian Schaipp](#), guest researcher, 2022 and 2023.
- [Slavomir Hanzely](#), intern, 2022; now Researcher at CISPA.
- [Aaron Mishkin](#), intern, 2023; now Postdoc at EPFL.
- [David Persson](#), Flatiron Fellow, 2024–present.
- [Michael Crawshaw](#), intern, 2025; joining Flatiron as a Fellow in 2026.
- [Tetiana Parshakova](#), Flatiron Fellow, 2025–present.

Recorded Talks

2026	The Polar Express. Instituto de Matematica Pura e Aplicada	recording/link
2024	New viewpoints, variants and convergence theory for stochastic Polyak step-sizes. MTL MLOpt	recording/link
2019	SGD: General Analysis and Improved Rates. ICML oral presentation	recording/link

Selected Invited Talks

Over 30 talks	in international conferences, seminars and workshops including.
2019	Parietal team seminar, Inria, Saclay International Conference on Continuous Optimization (ICCOPT), invited session, Berlin. Journée Statistique et Apprentissage, L’Institut des Hautes Études Scientifiques.
2018	The 21st International Conference on Artificial Intelligence and Statistics (AISTATS), semi plenary presentation and two posters, Lanzarote, Canary Islands Séminaire Parisien d’Optimisation, Institut Henri Poincaré International Symposium of Mathematical Optimization, invited session on Fast Converging Stochastic Optimization Algorithms, Bordeaux Seminar for the Machine Learning and Optimization group, EPFL

- 2017** [PGMO days](#), EDF'Lab Paris-Saclay
 Thirty-first Conference on Neural Information Processing Systems (Neurips), Poster, Vancouver.
[Optimization 2017](#), Faculdade de Ciências of the Universidade de Lisboa, invited talk, Lisboa
[The 27th Biennial Numerical Analysis Conference](#), Randomized Iterative Methods for Computing the Pseudoinverse, University of Strathclyde, Glasgow
[18th IMA Leslie Fox Prize talk in Numerical Analysis](#), awarded 2nd place, University of Strathclyde, Glasgow
[France/Japan Machine Learning Workshop](#), École Normale Supérieure, Paris
[Optimization, machine learning, and pluri-disciplinarity workshop](#), Paris, Inria Grenoble Rhone-Alpes, September 21-22.
 Probabilistic Numerics seminar, Max-Planck-Institute Tübingen
 Operations Research seminar, Center for Operations Research and Econometrics, Université catholique de Louvain
- 2016** Computational Mathematics and Applications seminar, Rutherford Appleton Laboratory
 Télécom ParisTech, Machine Learning and Statistics seminar, Paris
 International Conference on Continuous Optimization, Tokyo
 5th IMA Conference on Numerical Linear Algebra and Optimization, University of Birmingham
 International Conference on Machine Learning (**ICML**), New York
 Cambridge Image Analysis group seminar, University of Cambridge
Randomized iterative methods for linear systems
 Inria, SIERRA team seminar, Paris, *Randomized iterative methods for linear systems and inverting matrices*
- 2015** [Distributed machine learning and optimization](#), Alan Turing Scoping Workshop, Edinburgh, UK
Distributed Randomized Iterative Methods for Linear Systems
 The International Symposium on Mathematical Programming, Pittsburgh, USA.
Action constrained quasi-Newton methods
 Optimization and Big Data 2015, Edinburgh, UK
Randomized iterative methods for linear systems and inverting matrices
- 2014** [Irish Applied Mathematics Research Students' Meeting](#), Galway, Ireland.
Hunting inverses of matrix fields
[Best Talk Prize]
[Postgraduate Mathematics Colloquium](#), Edinburgh, UK
The history of optimization in blood and booze
 Cambridge Mathematics Society Research in the UK Afternoon, Cambridge, UK
Unconstrained optimization methods
[European Workshop on Advances in Continuous Optimization](#), Perpignan.
Generalizations of the quasi-Newton methods using an image constraint
- 2013** [International Conference on Continuous Optimization](#), Lisbon, Portugal
Third order methods and third order derivatives
 EURO-INFORMS Joint International Meeting, Rome, Italy
Third order methods using slices of the tensor and AD developments
- 2011** 28th Brazilian Colloquium on Mathematics, IMPA, Rio de Janeiro, Brazil
Automatic differentiation of Hessian matrices

Teaching

I have given over 360 hours of lectures on optimization, numerical analysis, and machine learning. Prior to that, as a graduate student at the University of Edinburgh, I gave over 350 hours of tutorials and lectures.

- 2019–2020: *Machine Learning in High Dimensions*, graduate course, Telecom Paris.
- 2017–2020: *Optimization and Numerical Analysis*, undergraduate course, Telecom Paris.
- 2017–2020: *Optimization for Data Science*, Data Science Master’s, Institut Polytechnique de Paris.
- 2019: *Stochastic Optimization for Machine Learning*, African Master’s of Machine Intelligence at AIMS, Rwanda.
- 2017–2019: Summer school lectures on stochastic optimization for machine learning at Yerevan State University and the University of Novi Sad.
- 2010–2015: Over 350 hours of tutorials and lectures at the University of Edinburgh, including linear algebra, calculus, financial mathematics, operations research, optimization theory, proofs, and problem solving.

Service

Editorial work and reviewing for:

Action Editor, *Transactions of Machine Learning Research* (TMLR)
Editorial Board Reviewers, *Journal of Machine Learning Research* (JMLR)
Area Chair / Meta-Reviewer, *International Conference on Machine Learning* (ICML), 2025
Area Chair, *Conference on Neural Information Processing Systems* (NeurIPS), 2021 and 2022
Senior Program Committee, *International Joint Conference on Artificial Intelligence* (IJCAI), 2020
Reviewer, *Conference on Neural Information Processing Systems* (NeurIPS)
International Conference on Machine Learning (ICML)
Journal of Machine Learning Research (JMLR)
Proceedings of the IEEE
IEEE Transactions on Signal Processing
SIAM Journal on Scientific Computing (SISC)
SIAM Journal on Optimization (SIOPT)
Optimization Methods and Software, Taylor & Francis
Computational and Applied Mathematics, Springer
Mathematical Programming, Springer Computation
BIT Numerical Mathematics, Springer
Numerical Algorithms, Springer

Publications

All entries below are generated directly from `../rob_papers.bib`.

- [1] N. Amsel, D. Persson, C. Musco, and R. M. Gower. “The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm”. In: *International Conference on Learning Representations (ICLR)*. Oral presentation and Outstanding Paper Honorable Mention. 2026. arXiv: [2505.16932](https://arxiv.org/abs/2505.16932) [cs.LG]. URL: <https://openreview.net/forum?id=yRtgZ1K8h0>.
- [2] M. Crawshaw, C. Modi, M. Liu, and R. M. Gower. “An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants”. In: *International Conference on Machine Learning (ICML)*. 2026. arXiv: [2510.09827](https://arxiv.org/abs/2510.09827) [cs.LG]. URL: <https://arxiv.org/abs/2510.09827>.
- [3] R. Islamov, M. Crawshaw, J. Cohen, and R. M. Gower. “Non-Euclidean Gradient Descent Operates at the Edge of Stability”. In: *International Conference on Machine Learning (ICML)*. Spotlight presentation. 2026. arXiv: [2603.05002](https://arxiv.org/abs/2603.05002) [cs.LG]. URL: <https://arxiv.org/abs/2603.05002>.

- [4] F. Schaipp, R. M. Gower, and A. B. Taylor. “Step-Size Stability in Stochastic Optimization: A Theoretical Perspective”. In: *International Conference on Machine Learning (ICML)*. 2026. arXiv: 2602.09842 [math.OC]. URL: <https://arxiv.org/abs/2602.09842>.
- [5] D. Cai, R. M. Gower, D. M. Blei, and L. K. Saul. “Fisher Meets Feynman: Score-Based Variational Inference with a Product of Experts”. In: *Conference on Neural Information Processing Systems (NeurIPS)*. Spotlight presentation. 2025. arXiv: 2510.21598 [stat.ML]. URL: <https://arxiv.org/abs/2510.21598>.
- [6] R. M. Gower, G. Garrigos, N. Loizou, D. Oikonomou, K. Mishchenko, and F. Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. 2025. arXiv: 2504.01898 [cs.LG]. URL: <https://arxiv.org/abs/2504.01898>.
- [7] A. Mishkin, A. Bietti, and R. M. Gower. “Level Set Teleportation: An Optimization Perspective”. In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*. Vol. 258. Proceedings of Machine Learning Research. PMLR, 2025, pp. 5059–5067. arXiv: 2403.03362. URL: <https://proceedings.mlr.press/v258/mishkin25a.html>.
- [8] A. Orvieto and R. M. Gower. “In Search of Adam’s Secret Sauce”. In: *Conference on Neural Information Processing Systems (NeurIPS)*. Oral presentation. 2025. arXiv: 2505.21829 [cs.LG]. URL: <https://openreview.net/forum?id=CH72XyZs4y>.
- [9] Y.-M. Pun, M. Buchholz, and R. M. Gower. *Schedulers for Schedule-free: Theoretically Inspired Hyperparameters*. 2025. arXiv: 2511.07767 [cs.LG]. URL: <https://arxiv.org/abs/2511.07767>.
- [10] F. Schaipp, G. Garrigos, U. Şimşekli, and R. M. Gower. “Tracking the Median of Gradients with a Stochastic Proximal Point Method”. In: *Transactions on Machine Learning Research (TMLR)* (2025). arXiv: 2402.12828 [cs.LG]. URL: <https://openreview.net/forum?id=WmxLEgYGxu>.
- [11] D. Cai, C. Modi, C. C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. “EigenVI: Score-Based Variational Inference with Orthogonal Function Expansions”. In: *Conference on Neural Information Processing Systems (NeurIPS)*. Vol. 37. Spotlight presentation. 2024. arXiv: 2410.24054. URL: <https://openreview.net/forum?id=thUf6ZB1Pp>.
- [12] D. Cai, C. Modi, L. Pillaud-Vivien, C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. “Batch and Match: Black-Box Variational Inference with a Score-Based Divergence”. In: *International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research. Spotlight presentation. PMLR, 2024, pp. 5258–5297. arXiv: 2402.14758. URL: <https://proceedings.mlr.press/v235/cai24b.html>.
- [13] R. M. Gower, D. A. Lorenz, and M. Winkler. “A Bregman-Kaczmarz Method for Nonlinear Systems of Equations”. In: *Computational Optimization and Applications (COAP)* 87.3 (2024), pp. 1059–1098. DOI: 10.1007/s10589-023-00541-9. arXiv: 2303.08549. URL: <https://doi.org/10.1007/s10589-023-00541-9>.
- [14] Y. Li, R. Yuan, C. Fan, M. Schmidt, S. Horváth, R. M. Gower, and M. Takáč. *Enhancing Policy Gradient with the Polyak Step-Size Adaptation*. 2024. arXiv: 2404.07525 [cs.LG]. URL: <https://arxiv.org/abs/2404.07525>.
- [15] A. Mishkin, A. Khaled, Y. Wang, A. Defazio, and R. M. Gower. “Directional Smoothness and Gradient Methods: Convergence and Adaptivity”. In: *Conference on Neural Information Processing Systems (NeurIPS)*. Vol. 37. 2024. arXiv: 2403.04081. URL: https://proceedings.neurips.cc/paper_files/paper/2024/hash/1ac83203e88eb6cf6b30642f0239b932-Abstract-Conference.html.
- [16] F. Schaipp, R. Ohana, M. Eickenberg, A. Defazio, and R. M. Gower. “MoMo: Momentum Models for Adaptive Learning Rates”. In: *International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research. PMLR, 2024, pp. 43542–43570. arXiv: 2305.07583. URL: <https://proceedings.mlr.press/v235/schaipp24a.html>.
- [17] B. Zhao, R. M. Gower, R. Walters, and R. Yu. “Improving Convergence and Generalization Using Parameter Symmetries”. In: *International Conference on Learning Representations (ICLR)*. 2024. arXiv: 2305.13404 [cs.LG]. URL: <https://openreview.net/forum?id=qXe7dE7Fq8M>.
- [18] J. Domke, R. M. Gower, and G. Garrigos. “Provable Convergence Guarantees for Black-Box Variational Inference”. In: *Conference on Neural Information Processing Systems (NeurIPS)*. Vol. 36. 2023. URL: https://proceedings.neurips.cc/paper_files/paper/2023/hash/d0bcff6425bbf850ec87d5327a965db9-Abstract-Conference.html.
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- [20] A. Khaled, O. Sebbouh, N. Loizou, R. M. Gower, and P. Richtárik. “Unified Analysis of Stochastic Gradient Methods for Composite Convex and Smooth Optimization”. In: *Journal of Optimization Theory and Applications (JOTA)* 199.2 (2023), pp. 499–540. DOI: [10.1007/s10957-023-02297-y](https://doi.org/10.1007/s10957-023-02297-y). arXiv: 2006.11573. URL: <https://doi.org/10.1007/s10957-023-02297-y>.
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- [26] J. Chen, R. Yuan, G. Garrigos, and R. M. Gower. “SAN: Stochastic Average Newton Algorithm for Minimizing Finite Sums”. In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*. Vol. 151. Proceedings of Machine Learning Research. PMLR, 2022, pp. 279–318. arXiv: 2106.10520. URL: <https://proceedings.mlr.press/v151/chen22b.html>.
- [27] N. Gazagnadou, M. Ibrahim, and R. M. Gower. “RidgeSketch: A Fast Sketching Based Solver for Large Scale Ridge Regression”. In: *SIAM Journal on Matrix Analysis and Applications (SIMAX)* 43.3 (2022), pp. 1440–1468. DOI: [10.1137/21M1422963](https://doi.org/10.1137/21M1422963). arXiv: 2001.01274. URL: <https://doi.org/10.1137/21M1422963>.
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- [29] Z. Wang, R. M. Gower, Y. Xia, L. He, and Y. Huang. “Randomized Iterative Methods for Low-Complexity Large-Scale MIMO Detection”. In: *IEEE Transactions on Signal Processing (TSP)* 70 (2022), pp. 2934–2949. DOI: [10.1109/TSP.2022.3180552](https://doi.org/10.1109/TSP.2022.3180552). URL: <https://doi.org/10.1109/TSP.2022.3180552>.
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